



UNIVERSITY OF MORATUWA

DEPARTMENT OF ELECTRONIC AND TELECOMMUNICATION ENGINEERING

**A portable multi sensor health monitor with on device
machine learning for real time arrhythmia and anomaly
detection**

Supervised by: Dr.Pranjeevan Kulasingham

Student Name : K.K.Jayani Nadeera

259103N

Project Overview

- This project aims to develop a **portable and real-time health monitoring device** that collects multiple physiological signals such as **ECG, heart rate, SpO₂, and body temperature** using integrated sensors. The collected data will be processed locally using **Edge/On Device Machine Learning** to detect **cardiac arrhythmias and abnormal health conditions** in real time.
The system is designed to provide **fast, continuous, and portable health monitoring** without relying entirely on cloud-based processing.

Novelty

- Secure Vital combines on-device AI for real time health anomaly detection with secure MQTT communication in a portable multi sensor device.

Project Objective

- **Develop a portable multi sensor health monitoring system using sensors such as:**
 - AD8232 for ECG
 - MAX30100 for heart rate and SpO₂
 - MLX90614 for body temperature
- **Implement lightweight machine learning on the ESP32** to classify ECG signals and detect abnormal cardiac patterns without depending continuously on cloud computing.
- **Reduce latency and internet dependency** by generating initial health alerts locally on the device, allowing detection to continue when Wi Fi or cloud connectivity is unavailable.
- **Implement secure MQTT communication** using lightweight authentication and encryption mechanisms to protect physiological data during transmission.
- **Evaluate the system** in terms of classification performance, response latency, resource consumption, communication security, and reliability under intermittent network conditions.

Methodology

1. Data acquisition & preprocessing

MIT BIH Arrhythmia Database + collected signals bandpass filter (0.5-100Hz), resample to 125Hz, normalize.

3. Training & evaluation

Grid search hyperparameters; evaluate with accuracy, precision, recall, F1-score and a confusion matrix.

5. Secure transmission

Publish sensor data and predictions over MQTT secured with TLS via WiFiClientSecure.

2. Model design

CNN with 3 convolutional layers + attention layer + 2 fully connected layers, following the reference architecture.

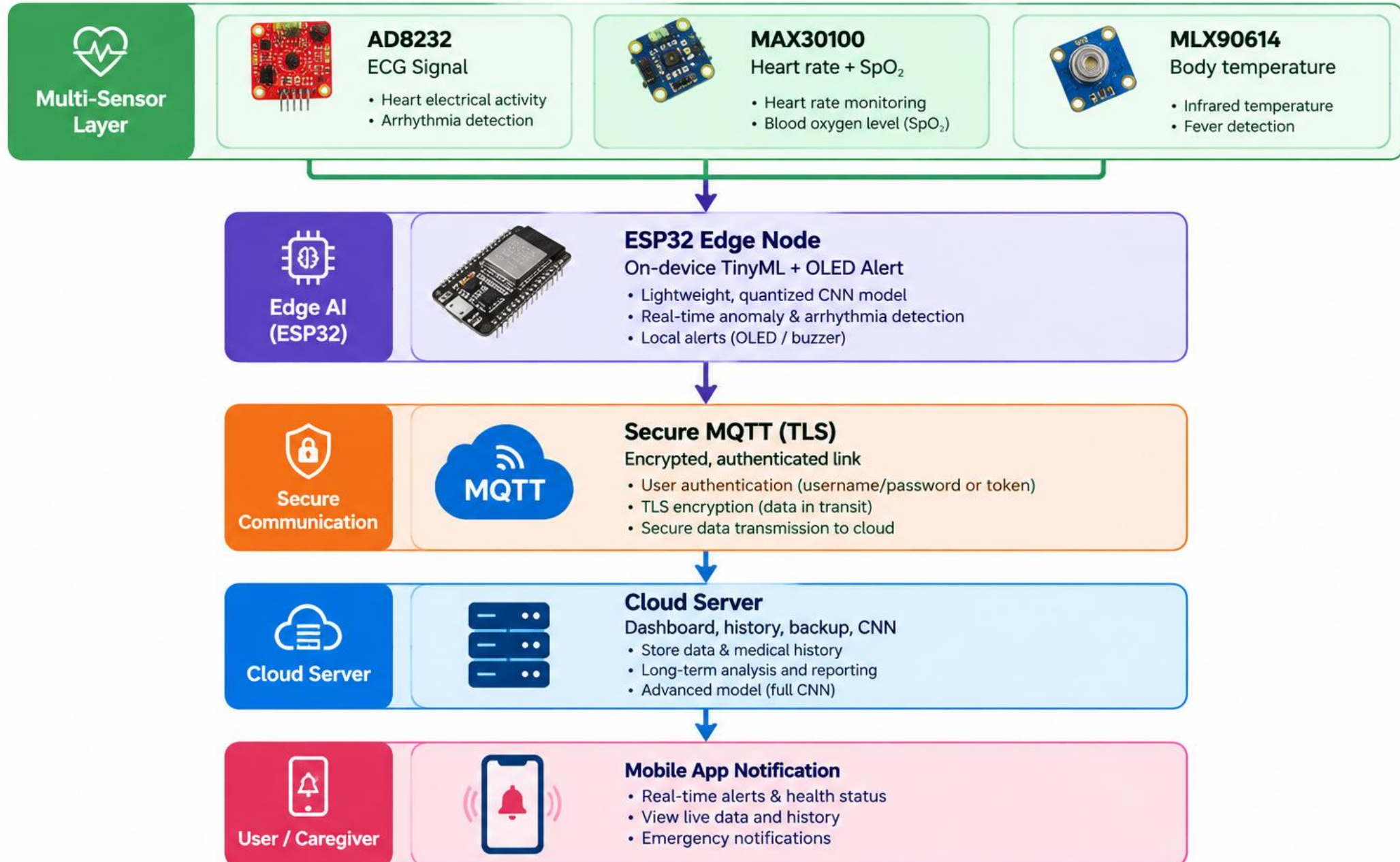
4. Edge deployment

Post training int8 quantization; convert to TensorFlow Lite Micro; flash onto the ESP32 for on device inference.

6. System integration & testing

Fuse sensor + edge ML + cloud dashboard; measure end to end alert .

SecureVital – System Architecture





SecureVital® – Multi-Sensor Specifications

SMART SENSING / BETTER HEALTH



AD8232 | ECG FRONT END



ECG Signal



2- or 3-Electrode Configuration

Function	Single-lead ECG / biopotential signal conditioning.
Supply voltage	2.0 – 3.5 V
Typical supply current	170 μ A
Gain	100
CMRR	80 dB from DC to 60 Hz
Electrodes	Supports 2- or 3-electrode configurations.
Filters	Adjustable 2-pole high-pass filter and 3-pole low-pass filter.
Lead-off detection	AC/DC lead-off detection.
RLD amplifier	Integrated right-leg-drive amplifier.
Output	Rail-to-rail output. Analog output for MCU ADC.
Package	20-lead, 4 mm x 4 mm LFCSP.
Use	ECG waveform acquisition, heart-rate and arrhythmia analysis.

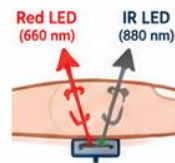


ECG

ECG waveform • Heart rate • Arrhythmia analysis



MAX30100 | PULSE OXIMETER + HEART RATE



Photodetector



Function	Optical heart-rate and SpO ₂ sensing.
Integrated components	Integrated red and infrared LEDs, photodetector, and low-noise analog front end.
Supply rails	1.8 V and 3.3 V
Sample rate / LED current	Programmable sample rate and LED current.
Shutdown current	0.7 μ A typical.
Performance	High SNR, ambient-light cancellation, high sample-rate capability, fast data output.
Interface	Digital I2C interface.
Package size	5.6 mm x 2.8 mm x 1.2 mm, 14-pin optically enhanced SiP.
Measurement concept	Finger → red/IR light absorption → heart rate and SpO ₂

i MAX30100 is obsolete; Analog Devices recommends MAX30101/MAX30102 for new designs.



Heart Rate + SpO₂

Pulse • Oxygen saturation • Wellness



MLX90614 | INFRARED TEMPERATURE



Infrared (non-contact)



Body / Object Temperature

Function	Non-contact infrared body/object temperature measurement.
Sensor temperature range	-40 to +125 °C
Object temperature range	-70 to +380 °C
Accuracy	±0.5 °C over 0 to +50 °C for both sensor and object temperature.
Resolution	0.02 °C
Supply versions	3.0 V and 5.0 V
Typical current	About 1.4 mA
Interface	Digital SMBus interface and configurable PWM output.
Internal components	Integrated thermopile detector, low-noise amplifier, 17-bit ADC and DSP.
Versions	Single- and dual-zone versions.
Package	TO-39 package.



Body Temperature

Non-contact • Accurate • Safe



ECG electrodes



AD8232

MCU ADC

ECG



Finger

MAX30100

I²C

MCU

Heart Rate + SpO₂



Body/skin

MLX90614

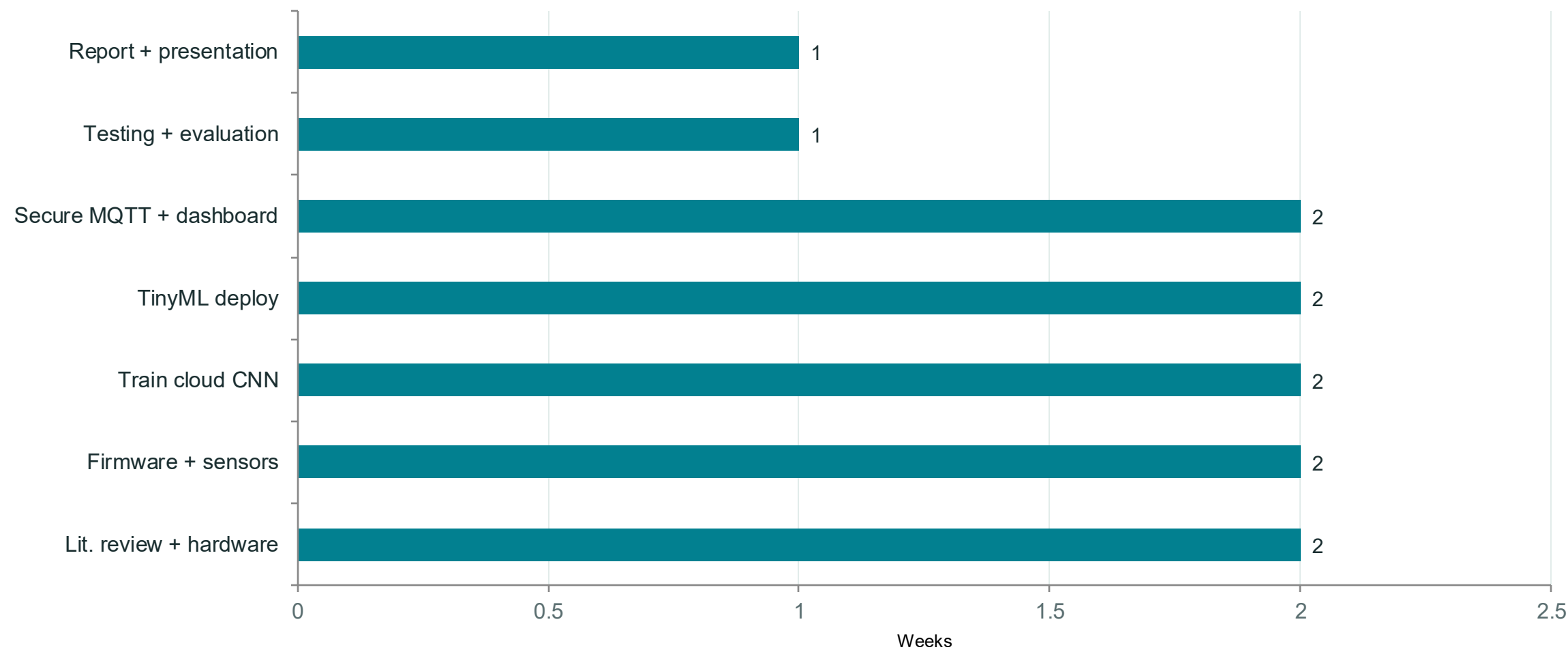
SMBus/PWM

MCU

Body Temperature

Milestone	Deliverables	Tasks
Milestone 1: Hardware & Edge-AI Development	D1.1 – Secure Vital PCB Prototype	T1.1: Design the complete circuit and PCB in Altium Designer , integrating ESP32, MAX30100, AD8232, MLX90614, OLED, power supply, and required interfaces.
	D1.2 – Working Edge-AI Prototype	T1.2: Interface and test all sensors, collect ECG/physiological data, develop and quantize the lightweight CNN/TinyML model, and implement real-time anomaly/arrhythmia detection on the ESP32.
Milestone 2: Security, Cloud & System Validation	D2.1 – Secure IoT Monitoring System	T2.1: Implement secure MQTT communication using authentication and TLS encryption, connect the ESP32 to the cloud dashboard, and implement data storage and caregiver/mobile alerts.
	D2.2 – Integrated SecureVital Prototype & Evaluation Report	T2.2: Integrate hardware, TinyML, security, cloud, and alerts; test accuracy, latency, power consumption, communication security, and operation during Wi-Fi interruption; document results and finalize the prototype.

Project timeline — 12 weeks



Expected outcomes

- A working portable prototype monitoring HR, SpO2, ECG and temperature
- On device arrhythmia classification comparable in accuracy to the cloud CNN baseline (target: within 2-3% of 98.2%)
- A secured MQTT pipeline

Thank you